Orange Condition Classification Using Classical Computer Vision and MobileNetV2

Green Pods Labs — Assignment 2

1. Objective

This project compares classical computer vision and deep learning approaches for classifying orange images into the dataset labels blackspot, canker, fresh, and grenning. The original spelling grenning is preserved.

Approach 1 uses CIELAB color histograms and Local Binary Pattern (LBP) texture features with SVM and Random Forest classifiers. Approach 2 uses MobileNetV2 transfer learning with ImageNet-pretrained weights.

2. Dataset and EDA

The uploaded dataset contains 1,090 images: 991 training images and 99 test images. Training counts are blackspot 184, canker 179, fresh 281, and grenning 347. Test counts are blackspot 22, canker 22, fresh 33, and grenning 22.

All 1,090 image files passed the Pillow integrity check in the Colab run. There were 150 unique image dimensions; 776 images were 800 × 800 pixels.

The class distribution is imbalanced: grenning is the largest class (369 total images, 33.9%) and canker is the smallest (201 total images, 18.4%). Macro precision, recall, and F1-score are therefore reported alongside accuracy.

3. Visual Bias and Data Quality

Visual inspection showed variation in lighting, reflections, shadows, image rotation, and backgrounds. Some rotated images contain black triangular areas around the fruit. A model could learn background or lighting shortcuts rather than only fruit condition.

Visible spots or surface damage are observations from the images, not verified biological diagnoses. Labels are treated as the dataset's ground-truth categories.

4. Preprocessing and Splits

The original training folder was split using stratified sampling into approximately 80% training (792 images) and 20% validation (199 images). The original 99-image test set was kept separate for final evaluation.

For classical ML, images were resized to 128 × 128. Sixteen-bin histograms were calculated for each of the three CIELAB channels, giving 48 color features. An LBP histogram with 10 bins was appended, giving 58 features in total based on the shown code.

For MobileNetV2, images were resized to 224 × 224 and normalized with ImageNet mean and standard deviation. Training-only augmentation used horizontal flips, small rotations, and mild color jitter.

5. Approach 1: Classical Computer Vision

CIELAB histograms summarize the distribution of lightness and chromatic color. LBP summarizes local grayscale texture patterns. These features were combined and passed to an RBF-kernel SVM with standardization and a Random Forest with 300 trees and balanced class weights.

6. Approach 2: MobileNetV2

MobileNetV2 was initialized with pretrained ImageNet weights. The feature extractor was frozen, and the final classifier was replaced with a four-class output layer. Training used cross-entropy loss and Adam for five epochs; the checkpoint with the lowest validation loss was retained.

Validation metrics were accuracy 0.9799, macro precision 0.9779, macro recall 0.9728, and macro F1 0.9748. The learning curves showed decreasing training and validation loss.

7. Evaluation Metrics

Accuracy is the fraction of all predictions that are correct. Precision measures the correctness of predictions for a class. Recall measures how many actual examples of a class are detected. Macro F1 averages the F1-score across classes equally and is useful when class counts differ.

8. Final Test Results

The held-out 99-image test confusion matrices showed SVM made 3 errors (96/99 correct), Random Forest made 3 errors (96/99 correct), and MobileNetV2 made 6 errors (93/99 correct).

9. Results Table

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
SVM	0.9697	0.9664	0.9659	0.9659
Random Forest	0.9697	0.9664	0.9659	0.9659
MobileNetV2	0.9394	0.9380	0.9318	0.9342

10. Inference Benchmark

On the Colab runtime reporting CUDA, prepared-input prediction timings were SVM 1.013 ms/image, Random Forest 80.953 ms/image, and MobileNetV2 11.161 ms/image. These exclude some image loading and preprocessing work and are not end-to-end production latency. Results depend on the runtime and should be repeated on target hardware.

11. Example Prediction

In the demonstrated test example, the actual label was blackspot. SVM predicted blackspot, Random Forest predicted blackspot, and MobileNetV2 predicted blackspot with a displayed confidence of 97.05%. One example does not guarantee general performance.

12. Mandatory Questions and Answers

Q1. In production, how does your model perform? On the 99-image test set, SVM and Random Forest each achieved 96.97% accuracy; MobileNetV2 achieved 93.94%. Prepared-input timings on the Colab CUDA runtime were 1.013, 80.953, and 11.161 ms/image, respectively. These are environment-specific measurements and not a production guarantee.

Q2. What are the drawbacks of your model? The test set is small, so metrics are uncertain. Blackspot and canker are confused. Lighting, glare, background, rotation, and class imbalance may affect predictions. Labels were not independently audited, and image appearance may not capture every relevant condition.

Q3. If given more time, what would you improve? Collect more balanced images across cameras and lighting; audit ambiguous labels; remove or mask backgrounds; evaluate on a larger independent dataset; use cross-validation and confidence intervals; tune features and hyperparameters; compare fine-tuned MobileNetV2 and other lightweight models; and benchmark end-to-end latency on target hardware.

Q4. Where did you spend the most time? The main work areas in this workflow were dataset inspection and preprocessing, building and debugging the classical feature pipeline, training and evaluating MobileNetV2, and debugging label-format consistency for confusion matrices. Replace this answer with your honest time breakdown if asked for exact percentages.

13. Conclusion

On this dataset split, SVM and Random Forest tied at 96.97% test accuracy and 0.9659 macro F1. MobileNetV2 achieved 93.94% test accuracy and 0.9342 macro F1. SVM is a reasonable candidate because it matched Random Forest accuracy and had the lowest measured prepared-input prediction time. The small test set means further independent evaluation is needed before production use.

14. Submission Checklist

Before submission, add the exact Colab notebook you ran, verify the feature matrix dimensions against the code, include model checkpoints if permitted, upload the repository to GitHub, configure the Colab link for anyone-with-link viewing, and record the required walkthrough. Do not claim these links or video exist until you have created them.